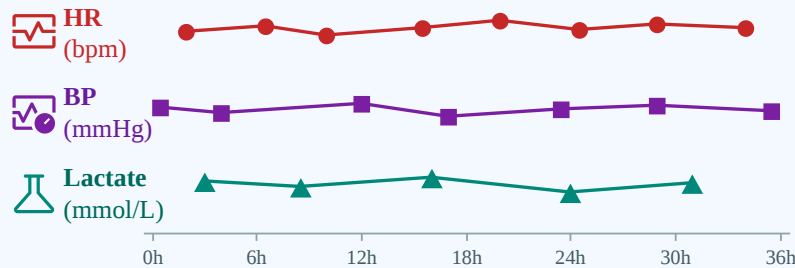


(a) TRIAGE Reasoning Procedure

Irregularly Sampled Medical Time Series (ISMTS)



Vitals

Labs

Static Features

text representation

Question

Will the patient experience in-hospital death during this ICU stay?

Patient: 67 years old, Male
HR: (2h, 100), (7h, 105), ...
Temp: (0h, 38.2), (3h, 38.5), ...
Glucose: (6h, 140), ...

LLM

Sequential reasoning chain

Step 1

Rationale for Survival

- improving oxygenation
- stable BP
- normal creatinine
- decreasing lactate
- ...

Step 2

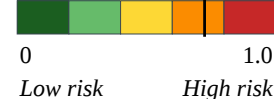
Rationale for Mortality

- high lactate
- persistent tachycardia
- worsening BP
- mechanical ventilation
- ...

Step 3

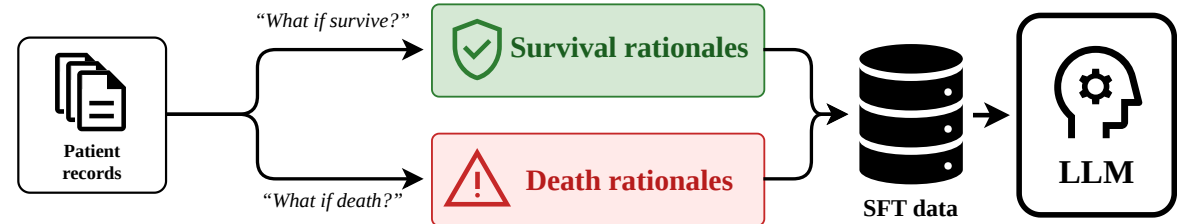
Risk Estimation

$$p(\text{"mortality"} \mid \text{input}) = 0.73$$
$$p(\text{"survival"} \mid \text{input}) = 0.27$$



(b) Two-stage Training Pipeline

Stage 1: Structured Reasoning Supervision



Stage 2: Self-Refinement

